SAYAK DUTTA

Github: github.com/sayakdutta | E-Mail ID: **sduttan598@gmail.com** | LinkedIn: **Sayak.Dutta**

Mobile Number: **(+49) 1706627596 (WhatsApp enabled)** | Address: **Germany | Open to Remote & Relocation**

Senior Machine Learning Engineer with 9+ years of experience designing and scaling production-grade AI systems. Specialized in **LLMs, RAG pipelines, real-time analytics, and full-stack AI platforms**, delivering end-to-end solutions that are **scalable, low-latency, and production-ready**. Skilled in **distributed infrastructure, MLOps pipelines, and intelligent applications** across Python, C++, TypeScript (Node.js, Next.js), Docker/Kubernetes, and modern cloud stacks. Thrive in **high-ownership, fast-paced environments** where rigorous engineering meets applied machine learning.

CORE COMPETENCIES

Programming: Python (Advanced), C++, TypeScript/JavaScript (Node.js, Next.js), PyTorch, LangChain, OpenAI API, Hugging Face
ML/AI Techniques: Time-series forecasting, NLP, recommendation engines, LLMs, deep learning, feature engineering, model validation
Data Engineering: ETL pipelines, time-series & structured data modeling, SQL (PostgreSQL)
Distributed Systems: Kafka, Redis, WebSockets, FAISS, Pinecone
Cloud & DevOps: Docker, Kubernetes, AWS (Lambda, S3), CI/CD (GitHub Actions)
ML Ops & Infra: Real-time ETL, model deployment, vector databases, pipeline orchestration, Google ADK
Other: FastAPI, Streamlit, Git/GitHub, Agile/Scrum, Open-Source contributions, Performance optimization

PROFESSIONAL EXPERIENCE

AI Engineer, **Biss Net GmbH**, Germany ***Nov 2025 – Present***

- Built a custom **multi-agent system** for ai.go using **Google ADK**, generating executable **Python code**, running it inside secure **Docker sandboxes**, and streaming real-time results with full guardrails and reproducibility.
- Optimized the **RAG pipeline** using **HNSW similarity search** and a **unified embedding client**, cutting retrieval latency and compute overhead while improving context consistency across large text corpora.
- Improved the platform's execution and retrieval stack with stronger **orchestration**, **caching layers**, and detailed **observability**, enabling faster reasoning cycles and more reliable end-to-end agent performance.

Senior Machine Learning Engineer, **Torto Labs Pvt Ltd.**, India (Remote) ***Aug 2021 – Sept 2025***

- Architected and deployed a distributed **RAG knowledge assistant (LangChain, FAISS, Redis)** supporting 100+ daily users, reducing lookup latency by 60%.
- Built core **ETL infrastructure (Python, Docker, AWS Lambda)** to process **1M+ records/month**, powering ML-driven forecasting.
- Streamlined **CI/CD pipelines** via **Docker** and **GitHub Actions**, achieving **95% reduction** in manual deployment time.
- Designed real-time **dashboards (Streamlit, PostgreSQL)** for analytics, accelerating decision-making by **35%**.
- Architected and deployed a large-scale, **PyTorch-based deep-learning platform** for **real-time video analytics**, processing over **1 million frames per hour** across hundreds of live camera streams with **sub-30 ms latency**, enabling **instant anomaly detection and automated responses** in production.

Machine Learning Engineer, **Berylls GmbH**, Germany ***Jul 2023 – Dec 2023***

- Developed and productionized real-time **recommendation engines** (**XGBoost**, collaborative filtering), increasing engagement by **25%**.
- Created scalable **NLP/ML pipelines** to classify **100K+ customer queries/month**, delivering **90% tagging accuracy**.
- Automated **ML deployment workflows (Docker, GitHub Actions)**, reducing operational overhead by **80%**.
- Collaborated with research/product teams for rapid delivery of new **AI features** in agile sprints.

Senior Data Analyst, **Nailbiter Inc.**, India ***Oct 2020 - Feb 2021***

- Engineered **text analytics pipelines** using **Hugging Face Transformers** for **200+ hours** of data, enabling real-time semantic search.
- Automated **API data ingestion (FastAPI, PostgreSQL)**, optimizing campaign targeting by **20%**.
- Shortened ideation-to-deployment cycle by **30%** via agile collaboration with cross-functional teams.

Analyst, **eClerx Services Ltd.,** India ***Apr 2018 – Oct 2020***

- Designed and integrated a **fuzzy-matching microservice (Python, Redis)** processing **500K+ records**, boosting data accuracy by **35%**.
- Implemented **real-time monitoring (Kafka)** to maintain **99.9% uptime** for distributed pipelines.
- **Built compliance dashboards (Power BI)** reducing manual data retrieval by 40%.

Project Manager, **HTL Aircon Pvt Ltd.,** India ***Dec 2016 – Mar 2018***

- **Led cross-functional teams** to deliver 10+ technical projects with 95% on-time completion.

SAYAK DUTTA

Github: github.com/sayakdutta | E-Mail ID: **sduttan598@gmail.com** | LinkedIn: **Sayak.Dutta**

Mobile Number: **(+49) 1706627596 (WhatsApp enabled)** | Address: **Germany | Open to Remote & Relocation**

- **Digitized reporting workflows** with Python, reducing processing time by 4 hours/week.

PROJECTS

ResearchRAG - AI Research Paper Digestor & Explorer

- Engineered a **full-stack RAG system** (FastAPI + FAISS) that processes 30-page academic PDFs into summaries, strengths/weaknesses, and future research directions in under 30 seconds.
- Built an **LLM-powered interactive chat** grounded in embeddings, delivering 95%+ context-relevant answers using free OpenRouter GPT-OSS models.
- Delivered a **production-ready solution** with robust error handling, logging, and seamless frontend–backend integration.
- Designed a **responsive UI** with exportoptions (PDF/Markdown), making research digestion 3–5× faster for researchers and students.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, FAISS, PyMuPDF, OpenRouter GPT-OSS.

AskPostgres - Natural Language to SQL for PostgreSQL

- Built a full-stack application that converts natural language into SQL queries, enabling **business users to query PostgreSQL databases without technical SQL knowledge**.
- Implemented authentication, role-based access control, error handling, and interactive UI with light/dark themes; validated with **extensive query variations** to ensure robustness.
- Improved analyst efficiency by **reducing SQL-writing time by up to 80%**, supporting business analytics, data exploration, and performance monitoring.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, FAISS, PyMuPDF, OpenRouter GPT-OSS.

OutreachPilot - Campaign Automation & Tracking Platform

- Engineered a full-stack automation platform for end-to-end campaign orchestration using **FastAPI**, **Next.js**, **Tailwind**, and **PostgreSQL**, ensuring smooth and reliable workflow execution.
- Integrated automated email delivery via the **Resend API**, with **custom pixel + redirect tracking** to accurately measure open rates, CTR, and variant performance.
- Built modular backend **microservices** with **async workers**, **queues**, and **real-time logging**, delivering high scalability, stability, and fault-tolerant performance.
- Delivered production-ready infrastructure with **Dockerized deployment** and **CI/CD pipelines**, enabling enterprise-grade reliability and rapid feature iteration.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, PostgreSQL, Docker, Resend API

TalkToTube - YouTube Transcript Explorer with LLM Q&Arer

- Built a **full-stack application** (FastAPI, Next.js, Tailwind, PostgreSQL, Docker) that ingests and processes YouTube transcripts into searchable, structured text.
- Implemented an **LLM-powered Q&A system** enabling users to ask natural language questions and receive context-grounded answers from video content.
- Designed a **responsive web interface** with role-based access, real-time results, and export options for summaries and highlights.
- Integrated **vector search (FAISS)** and embeddings for efficient retrieval, delivering fast and accurate results across long-form video content.
- Delivered production-ready infra with **containerized deployment (Docker)** and CI/CD pipelines, ensuring scalability and reproducibility.

Tech stack: FastAPI, Next.js, Tailwind, TypeScript, PostgreSQL, FAISS, Docker

EDUCATION

University of Mumbai, Mumbai, India — ***August 2012 – May 2016***

Bachelor of Engineering– Mechanical Engineering | Grade: First Division

Languages: English (Fluent), Hindi (Native), German (A2, actively improving)

Visa: Valid in Germany; open to relocation **Work Authorization:** EU Blue Card eligible | Open to employer-sponsored visas (US/UK)